Chapter 1
Introduction

The ecosystem of Earth's oceanic environments is difficult to understand with our current technology. The lack of light beneath the surface of the ocean necessities the use of acoustic based analysis. Analysis of marine presence requires technology, knowledge, and time. In particular, auditory analysis and identification requires manual review of sound data to make annotations of animal presence. In this thesis we use machine learning methods to automatically process audio recordings, reducing the tedium of the annotation process.

This study focuses on the humpback whales, known for their acoustic presence beneath the ocean surface. Whale songs are complex sounds primarily used by males (cite) to communicate with other whales, with the purposes of attracting mates or establishing social dominance. Songs vary in length, frequency, inflections, and more. Reasons for this variation include but are not limited to whale species, geographic location, and the individual.

The data in this work was generated by project ######. In this study a hydrophone was dropped in [location] for [periods of time] over [date range].... A hydrophone is a type of microphone built to withstand the high pressure of deep water environments. This is a type of non-invasive data collection technique known as passive acoustic monitoring (PAM). PAM allows for the collection of long term sound data without the need for a human operator. After the desired amount of recording has passed, the hydrophone is removed from the water. A group of Cal Poly [marine science students/Maddie] analyzed the data by simultaneously listening and inspecting the spectrogram. The analysis and annotation process is time consuming. The amount of time that analysts must spend listening to the recording is, at minimum, the amount of time the hydrophone is kept in the water. Unfortunately this process does not accelerate if there is a significant amount of inactivity. Skipping portions of the recording without additional information runs the risk of skipping over important whale call information.

Machine learning models aim to reduce time for manual annotation. These models detect patterns and key information that differentiate whale calls from other noise in manually annotated data. This information informs the educated decisions about the presence of whale calls when presented with new sound data. The additional analysis provided by these models provides a layer of confidence for the analysts. Machine learning models can detect periods of downtime, and help annotators skip directly to the next sound of interest.

However, predicting on sound data is more difficult due to the time nature of data. Typical inputs for machine learning data are independent rows with several explanatory variables and a single response variable of interest. Due to the nature of the data, a whale call can not be treated as an observational unit. Whale calls differ in length, and sometimes whale calls may overlap. To address these issues, we discretize time into intervals. If a whale call occurs in a time interval, then the observation is positive. If no whale call occurs in the interval, then the observation is positive.

This thesis focuses on the application of machine learning models such as k-nearest-neighbors and random forests on discretized sound data to detect humpback whale calls off the central coast of California. The primary goal is to avoid periods of inactivity while not overlooking humpback whale songs.

Chapter 2 of this thesis provides an overview of other literature focused on identifying whales throughout the world. Chapter 3 details the methods used to transform the continuous sound data into a model used for machine learning. This includes the transformation of the sound data, additional pre-processing changes, and post analysis aggregation. Chapter 4 discusses the results of some of the analyses. Chapter 5 concludes with final thoughts, limitations in the analysis, and areas for future work.